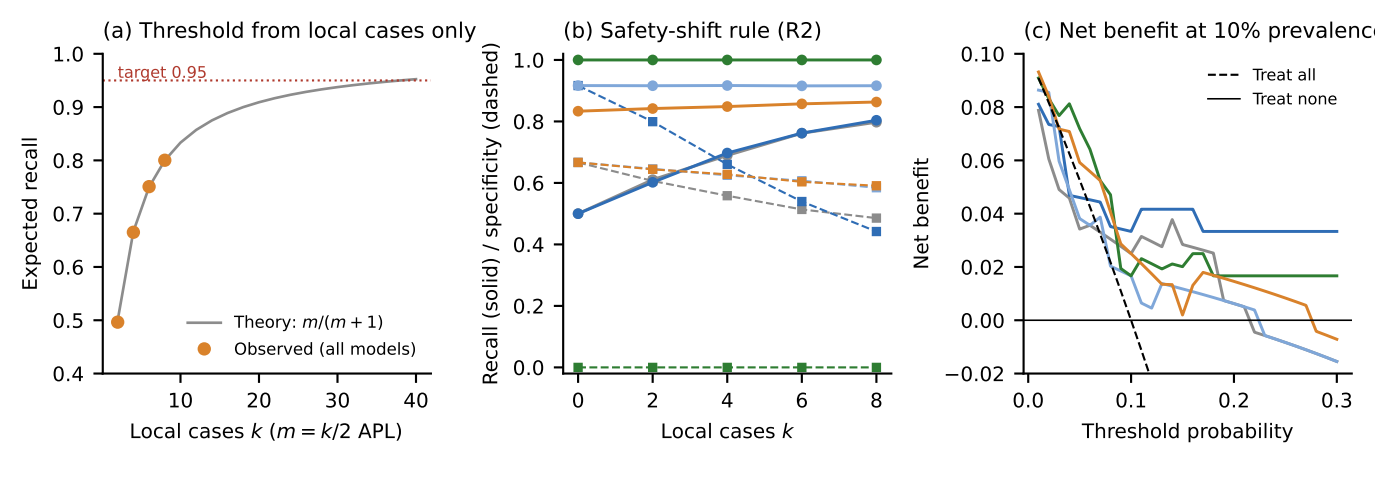
(a) Threshold from local cases only
target 0.95
Expected recall
Local cases k ($m = k/2$ APL)
Theory: $m/(m+1)$
Observed (all models)
(b) Safety-shift rule (R2)
Recall (solid) / specificity (dashed)
Local cases k
(c) Net benefit at 10% prevalence
Treat all
Treat none
Net benefit
Threshold probability
EfficientNet-B0, no PCA
EfficientNet-B0 + PCA-64
EfficientNet-B0 + PCA-16
Engineered, attention MIL
Fine-tuned EfficientNet-B